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Knowledge-Based Systems (KBS)

Certainty Factor (CF) Diagnostic Report

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1. Methodology & Logical Framework

This report follows the rigorous 3-step methodology for reasoning under uncertainty:

- Step 1: Evidence Logic:** For rules with multiple conditions joined by **AND**, take the minimum certainty: $CF(Evidence) = \min(CF_{E1}, CF_{E2}, \dots, CF_{En})$.
- Step 2: Apply Rule:** Calculate the certainty of the hypothesis (H) given the evidence (E): $CF(H, E) = CF(Evidence) \times CF(Rule)$.
- Step 3: Combine Certainty Factors:** When multiple rules support the same hypothesis, combine them: $CF_{combined} = CF_{old} + CF_{new}(1 - CF_{old})$.

Important Academic Note on Semantic Grouping

“Related hypotheses referring to the same root mechanical failure were grouped semantically before CF combination.”

Consistency Standards

- Rounding:** All final Certainty Factors are rounded to **4 decimal places**.
 - Constants:** $CF(Rule) = 0.7$; $CF(Evidence) = \text{Randomly simulated } (0.5 - 0.9)$.
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2. Final Simulation Summary

🏆 Highest Certainty Diagnosis: Alternator Failure

📊 Confidence Level: 0.8939

Top 5 Diagnoses Comparison

Rank	Diagnosis	Final CF
1	Alternator Failure	0.8939
2	ABS Sensor / Module Failure	0.8870
3	radiator_strangled	0.5880
4	Head Gasket Breach	0.5866
5	brake_fluid_escaping	0.5810

3. Detailed Diagnostic Calculations (Step-by-Step)

Diagnosis: Head Gasket Breach

Report Note: The system combines these because Rule 68 provides more specific ‘conspiracy’ evidence than Rule 46, strengthening the final verdict.

Rule 1: open_case(head_gasket_breached)

Statement: IF smoke_color = white AND coolant_loss = true THEN open_case(head_gasket_breached)

Calculation Steps:

- Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.59, 0.51) = 0.51$
- Step 2 (Apply Rule):** $CF(H,E) = 0.51 \times 0.7 = 0.3570$

Rule 2: close_case(head_gasket_confirmed_guilty)

Statement: IF temp > 110C AND smoke_color = white AND coolant_loss = true THEN close_case(head_gasket_confirmed_guilty) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.54, 0.59, 0.51) = 0.51$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.51 \times 0.7 = 0.3570$

Step 3: Combination for Head Gasket Breach

1. Initial CF: **0.3570**
2. Combine with Rule 2 (CF=0.3570):
 - $CF_{\text{combined}} = 0.3570 + 0.3570(1 - 0.3570) = 0.5866$

Final Combined Certainty for Head Gasket Breach: 0.5866

Diagnosis: Alternator Failure

Report Note: By combining these, the system distinguishes between a simple 'old battery' and an alternator that is actively failing and affecting engine performance.

Rule 1: open_case(alternator_refusing_to_work)

Statement: IF battery_light = ON AND alternator_voltage < 13.5V THEN open_case(alternator_refusing_to_work) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.65, 0.58) = 0.58$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.58 \times 0.7 = 0.4060$

Rule 2: open_case(charging_system_failing_in_darkness)

Statement: IF headlights_dim = true AND alternator_voltage < 13V THEN open_case(charging_system_failing_in_darkness) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.57, 0.65) = 0.57$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.57 \times 0.7 = 0.3990$

Rule 3: open_case(alternator_sabotaging_engine_management)

Statement: IF engine_light = ON AND battery_light = ON AND rpm_unstable = true THEN open_case(alternator_sabotaging_engine_management) **Calculation**

Steps:

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.65, 0.76) = 0.65$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.65 \times 0.7 = \mathbf{0.4550}$

Rule 4: close_case(full_charging_system_collapse)

Statement: IF battery_drain = true AND slow_crank = true AND alternator_voltage < 13V THEN close_case(full_charging_system_collapse)

Calculation Steps:

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.67, 0.68, 0.65) = 0.65$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.65 \times 0.7 = \mathbf{0.4550}$

Step 3: Combination for Alternator Failure

1. Initial CF: **0.4060**
2. Combine with Rule 2 (CF=0.3990):
 - $CF_{\text{combined}} = 0.4060 + 0.3990(1 - 0.4060) = 0.6430$
3. Combine with Rule 3 (CF=0.4550):
 - $CF_{\text{combined}} = 0.6430 + 0.4550(1 - 0.6430) = 0.8054$
4. Combine with Rule 4 (CF=0.4550):
 - $CF_{\text{combined}} = 0.8054 + 0.4550(1 - 0.8054) = 0.8939$

Final Combined Certainty for Alternator Failure: 0.8939

Diagnosis: ABS Sensor / Module Failure

Report Note: The system merges evidence from the Braking, Traction, and Wheel-speed logic to confirm a high-confidence sensor failure.

Rule 1: open_case(ABS_sensor_gone_silent)

Statement: IF ABS_light = ON AND wheel_speed_signal = missing THEN open_case(ABS_sensor_gone_silent) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.61, 0.75) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = \mathbf{0.4270}$

Rule 2: open_case(ABS_brain_dead)

Statement: IF ABS_light = ON AND module_response = false THEN open_case(ABS_brain_dead) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.61, 0.65) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = \mathbf{0.4270}$

Rule 3: open_case(shared_sensor_taking_down_two_systems)

Statement: IF brake_light = ON AND ABS_light = ON AND wheel_speed_signal = missing THEN open_case(shared_sensor_taking_down_two_systems) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.83, 0.61, 0.75) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = \mathbf{0.4270}$

Rule 4: close_case(single_sensor_paralyzed_two_systems)

Statement: IF ABS_fault = true AND traction_control_fault = true AND wheel_speed_missing = true THEN close_case(single_sensor_paralyzed_two_systems) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.7, 0.57, 0.87) = 0.57$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.57 \times 0.7 = \mathbf{0.3990}$

Step 3: Combination for ABS Sensor / Module Failure

1. Initial CF: **0.4270**
2. Combine with Rule 2 (CF=0.4270):
 - $CF_{\text{combined}} = 0.4270 + 0.4270(1 - 0.4270) = \mathbf{0.6717}$

3. Combine with Rule 3 (CF=0.4270):

- $CF_{combined} = 0.6717 + 0.4270(1 - 0.6717) = 0.8119$

4. Combine with Rule 4 (CF=0.3990):

- $CF_{combined} = 0.8119 + 0.3990(1 - 0.8119) = 0.8870$

Final Combined Certainty for ABS Sensor / Module Failure: 0.8870

Diagnosis: fuel_injector_suspect

Statement: IF engine_light = ON AND rpm_drop = true THEN
open_case(fuel_injector_suspect) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.51) = 0.51$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.51 \times 0.7 = 0.3570$ **Final Certainty:** 0.3570

Diagnosis: vacuum_leak_suspect

Statement: IF engine_light = ON AND fuel_trim > 15% THEN
open_case(vacuum_leak_suspect) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = 0.4270$ **Final Certainty:** 0.4270

Diagnosis: dirty_MAF_sensor

Statement: IF engine_light = ON AND maf_reading < threshold THEN
open_case(dirty_MAF_sensor) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.59) = 0.59$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.59 \times 0.7 = 0.4130$ **Final Certainty:** 0.4130

Diagnosis: throttle_body_obstruction

Statement: IF engine_stall = true AND idle_rpm < 500 THEN
open_case(throttle_body_obstruction) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.79, 0.77) = 0.77$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.77 \times 0.7 = 0.5390$ **Final Certainty:** 0.5390

Diagnosis: fuel_pump_dying

Statement: IF engine_stall = true AND fuel_pressure < 30psi THEN open_case(fuel_pump_dying) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.79, 0.86) = 0.79$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.79 \times 0.7 = 0.5530$ **Final Certainty:** 0.5530

Diagnosis: ignition_system_breakdown

Statement: IF misfire_detected = true AND spark_plug_gap > 1.6mm THEN open_case(ignition_system_breakdown) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.53, 0.67) = 0.53$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.53 \times 0.7 = 0.3710$ **Final Certainty:** 0.3710

Diagnosis: clogged_injector

Statement: IF misfire_detected = true AND injector_pulse_weak = true THEN open_case(clogged_injector) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.53, 0.51) = 0.51$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.51 \times 0.7 = 0.3570$ **Final Certainty:** 0.3570

Diagnosis: coolant_escape

Statement: IF engine_temp > 110C AND coolant_level < min THEN open_case(coolant_escape) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.59, 0.7) = 0.59$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.59 \times 0.7 = 0.4130$ **Final Certainty:** 0.4130

Diagnosis: cooling_fan_dead

Statement: IF engine_temp > 110C AND radiator_fan_rpm = 0 THEN open_case(cooling_fan_dead) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.59, 0.51) = 0.51$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.51 \times 0.7 = 0.3570$ **Final Certainty:** 0.3570

Diagnosis: thermostat_frozen_shut

Statement: IF engine_temp > 110C AND thermostat_open = false THEN open_case(thermostat_frozen_shut) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.59, 0.58) = 0.58$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.58 \times 0.7 = \mathbf{0.4060}$ **Final Certainty:** 0.4060

Diagnosis: oil_bleeding_out

Statement: IF oil_pressure < 10psi AND oil_level < min THEN open_case(oil_bleeding_out) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.72) = 0.72$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.72 \times 0.7 = \mathbf{0.5040}$ **Final Certainty:** 0.5040

Diagnosis: oil_pump_cardiac_arrest

Statement: IF oil_pressure < 10psi AND oil_level = normal THEN open_case(oil_pump_cardiac_arrest) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.59) = 0.59$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.59 \times 0.7 = \mathbf{0.4130}$ **Final Certainty:** 0.4130

Diagnosis: lubrication_starvation

Statement: IF knocking_sound = true AND oil_viscosity_wrong = true THEN open_case(lubrication_starvation) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.74, 0.82) = 0.74$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.74 \times 0.7 = \mathbf{0.5180}$ **Final Certainty:** 0.5180

Diagnosis: timing_chain_jumped

Statement: IF knocking_sound = true AND timing_advance_off = true THEN open_case(timing_chain_jumped) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.74, 0.5) = 0.50$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.50 \times 0.7 = \mathbf{0.3500}$ **Final Certainty:** 0.3500

Diagnosis: fuel_delivery_dead_on_arrival

Statement: IF crank = true AND start = false AND fuel_pressure = 0 THEN open_case(fuel_delivery_dead_on_arrival) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.82, 0.78, 0.64) = 0.64$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.64 \times 0.7 = 0.4480$ **Final Certainty:** 0.4480

Diagnosis: fluid_hemorrhage

Statement: IF gear_slip = true AND fluid_level < min THEN open_case(fluid_hemorrhage) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.56, 0.88) = 0.56$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.56 \times 0.7 = 0.3920$ **Final Certainty:** 0.3920

Diagnosis: clutch_pack_disintegration

Statement: IF gear_slip = true AND fluid_level = normal THEN open_case(clutch_pack_disintegration) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.56, 0.63) = 0.56$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.56 \times 0.7 = 0.3920$ **Final Certainty:** 0.3920

Diagnosis: fluid_burned_beyond_recovery

Statement: IF harsh_shift = true AND fluid_color = black THEN open_case(fluid_burned_beyond_recovery) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.54, 0.54) = 0.54$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.54 \times 0.7 = 0.3780$ **Final Certainty:** 0.3780

Diagnosis: shift_solenoid_unresponsive

Statement: IF harsh_shift = true AND solenoid_response = false THEN open_case(shift_solenoid_unresponsive) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.54, 0.84) = 0.54$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.54 \times 0.7 = 0.3780$ **Final Certainty:** 0.3780

Diagnosis: thermal_meltdown

Statement: IF transmission_temp > 130C AND load = towing THEN
open_case(thermal_meltdown) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.74, 0.82) = 0.74$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.74 \times 0.7 = 0.5180$ **Final Certainty:** 0.5180

Diagnosis: transmission_cooler_strangled

Statement: IF transmission_temp > 130C AND cooler_flow = blocked THEN
open_case(transmission_cooler_strangled) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.74, 0.79) = 0.74$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.74 \times 0.7 = 0.5180$ **Final Certainty:** 0.5180

Diagnosis: inhibitor_switch_lying

Statement: IF no_drive_engagement = true AND neutral_switch = fault THEN
open_case(inhibitor_switch_lying) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.71, 0.89) = 0.71$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.71 \times 0.7 = 0.4970$ **Final Certainty:** 0.4970

Diagnosis: CV_joint_self_destructing

Statement: IF vibration_on_acceleration = true AND cv_boot_torn = true THEN
open_case(CV_joint_self_destructing) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.65, 0.72) = 0.65$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.65 \times 0.7 = 0.4550$ **Final Certainty:** 0.4550

Diagnosis: brake_fluid_escaping

Statement: IF brake_light = ON AND fluid_level < min THEN
open_case(brake_fluid_escaping) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.83, 0.88) = 0.83$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.83 \times 0.7 = 0.5810$ **Final Certainty:** 0.5810

Diagnosis: pads_ground_to_metal

Statement: IF brake_light = ON AND pad_sensor_triggered = true THEN open_case(pads_ground_to_metal) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.83, 0.75) = 0.75$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.75 \times 0.7 = 0.5250$ **Final Certainty:** 0.5250

Diagnosis: air_infiltration

Statement: IF pedal_travel_excessive = true AND air_in_lines = true THEN open_case(air_infiltration) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.84, 0.73) = 0.73$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.73 \times 0.7 = 0.5110$ **Final Certainty:** 0.5110

Diagnosis: master_cylinder_surrender

Statement: IF pedal_travel_excessive = true AND master_cylinder_bypass = true THEN open_case(master_cylinder_surrender) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.84, 0.78) = 0.78$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.78 \times 0.7 = 0.5460$ **Final Certainty:** 0.5460

Diagnosis: rotor_warped_under_stress

Statement: IF brake_vibration = true AND rotor_runout > 0.05mm THEN open_case(rotor_warped_under_stress) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.52, 0.59) = 0.52$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.52 \times 0.7 = 0.3640$ **Final Certainty:** 0.3640

Diagnosis: left_caliper_seized_hostage

Statement: IF pull_left = true AND left_caliper_pressure = 0 THEN open_case(left_caliper_seized_hostage) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.62, 0.53) = 0.53$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.53 \times 0.7 = 0.3710$ **Final Certainty:** 0.3710

Diagnosis: right_caliper_seized_hostage

Statement: IF pull_right = true AND right_caliper_pressure = 0 THEN open_case(right_caliper_seized_hostage) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.59, 0.54) = 0.54$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.54 \times 0.7 = 0.3780$ **Final Certainty:** 0.3780

Diagnosis: battery_exhausted

Statement: IF battery_light = ON AND battery_voltage < 12V THEN open_case(battery_exhausted) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.65, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = 0.4270$ **Final Certainty:** 0.4270

Diagnosis: hidden_electrical_thief

Statement: IF battery_drain_overnight = true AND parasitic_draw > 50mA THEN open_case(hidden_electrical_thief) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.87, 0.76) = 0.76$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.76 \times 0.7 = 0.5320$ **Final Certainty:** 0.5320

Diagnosis: battery_too_weak_to_fight

Statement: IF starter_single_click = true AND battery_voltage < 12V THEN open_case(battery_too_weak_to_fight) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.74, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = 0.4270$ **Final Certainty:** 0.4270

Diagnosis: starter_relay_gone_rogue

Statement: IF starter_no_response = true AND relay_signal = false THEN open_case(starter_relay_gone_rogue) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.57, 0.79) = 0.57$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.57 \times 0.7 = 0.3990$ **Final Certainty:** 0.3990

Diagnosis: wiring_short_in_hiding

Statement: IF fuse_blows_repeatedly = true AND resistance_drop_detected = true THEN open_case(wiring_short_in_hiding) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.9, 0.76) = 0.76$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.76 \times 0.7 = 0.5320$ **Final Certainty:** 0.5320

Diagnosis: O2_sensor_gone_blind

Statement: IF check_engine = ON AND O2_sensor_voltage_flat = true THEN open_case(O2_sensor_gone_blind) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.72, 0.77) = 0.72$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.72 \times 0.7 = 0.5040$ **Final Certainty:** 0.5040

Diagnosis: catalytic_converter_defeated

Statement: IF check_engine = ON AND catalyst_efficiency < 80% THEN open_case(catalytic_converter_defeated) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.72, 0.84) = 0.72$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.72 \times 0.7 = 0.5040$ **Final Certainty:** 0.5040

Diagnosis: engine_drowning_in_fuel

Statement: IF smoke_color = black THEN open_case(engine_drowning_in_fuel) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.81) = 0.81$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.81 \times 0.7 = 0.5670$ **Final Certainty:** 0.5670

Diagnosis: piston_rings_surrendered

Statement: IF smoke_color = blue AND oil_consumption_high = true THEN open_case(piston_rings_surrendered) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.63, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = 0.4270$ **Final Certainty:** 0.4270

Diagnosis: EGR_valve_frozen_in_crime

Statement: IF exhaust_smell = strong AND EGR_position = stuck_open THEN open_case(EGR_valve_frozen_in_crime) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.58, 0.88) = 0.58$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.58 \times 0.7 = \mathbf{0.4060}$ **Final Certainty:** 0.4060

Diagnosis: unburned_fuel_escaping_justice

Statement: IF HC_emissions > limit AND misfire_count > 0 THEN open_case(unburned_fuel_escaping_justice) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.85, 0.63) = 0.63$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.63 \times 0.7 = \mathbf{0.4410}$ **Final Certainty:** 0.4410

Diagnosis: tires_fighting_each_other

Statement: IF steering_pull = true AND tire_pressure_diff > 5psi THEN open_case(tires_fighting_each_other) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.66) = 0.66$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.66 \times 0.7 = \mathbf{0.4620}$ **Final Certainty:** 0.4620

Diagnosis: alignment_corrupted

Statement: IF steering_pull = true AND camber_diff > 1 degree THEN open_case(alignment_corrupted) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.87) = 0.76$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.76 \times 0.7 = \mathbf{0.5320}$ **Final Certainty:** 0.5320

Diagnosis: wheel_imbalance_tremors

Statement: IF steering_vibration = true AND wheel_balance_off > 10g THEN open_case(wheel_imbalance_tremors) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.68, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = \mathbf{0.4270}$ **Final Certainty:** 0.4270

Diagnosis: power_steering_bleeding_out

Statement: IF steering_effort_high = true AND PS_fluid < min THEN open_case(power_steering_bleeding_out) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.6, 0.72) = 0.60$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.60 \times 0.7 = \mathbf{0.4200}$ **Final Certainty:** 0.4200

Diagnosis: PS_pump_giving_up

Statement: IF steering_effort_high = true AND PS_pump_pressure < threshold THEN open_case(PS_pump_giving_up) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.6, 0.61) = 0.60$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.60 \times 0.7 = \mathbf{0.4200}$ **Final Certainty:** 0.4200

Diagnosis: sway_bar_link_broken_loose

Statement: IF clunk_on_bump = true AND sway_bar_link_play > 5mm THEN open_case(sway_bar_link_broken_loose) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.73, 0.86) = 0.73$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.73 \times 0.7 = \mathbf{0.5110}$ **Final Certainty:** 0.5110

Diagnosis: shock_absorber_leaking_evidence

Statement: IF clunk_on_bump = true AND shock_absorber_leak = true THEN open_case(shock_absorber_leaking_evidence) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.73, 0.66) = 0.66$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.66 \times 0.7 = \mathbf{0.4620}$ **Final Certainty:** 0.4620

Diagnosis: suspension_geometry_corrupted

Statement: IF uneven_tire_wear = true AND camber_out_of_spec = true THEN open_case(suspension_geometry_corrupted) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.59, 0.9) = 0.59$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.59 \times 0.7 = \mathbf{0.4130}$ **Final Certainty:** 0.4130

Diagnosis: fuel_line_hemorrhaging

Statement: IF fuel_smell_cabin = true AND fuel_line_wet = true THEN open_case(fuel_line_hemorrhaging) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.7, 0.54) = 0.54$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.54 \times 0.7 = \mathbf{0.3780}$ **Final Certainty:** 0.3780

Diagnosis: engine_suffocating

Statement: IF mpg_drop > 20% AND air_filter_restriction > 50% THEN open_case(engine_suffocating) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.52, 0.54) = 0.52$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.52 \times 0.7 = \mathbf{0.3640}$ **Final Certainty:** 0.3640

Diagnosis: injectors_starving_engine

Statement: IF mpg_drop > 20% AND injector_flow_rate_low = true THEN open_case(injectors_starving_engine) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.52, 0.75) = 0.52$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.52 \times 0.7 = \mathbf{0.3640}$ **Final Certainty:** 0.3640

Diagnosis: fuel_pressure_regulator_erratic

Statement: IF engine_surge = true AND fuel_pressure_fluctuates = true THEN open_case(fuel_pressure_regulator_erratic) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.82, 0.67) = 0.67$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.67 \times 0.7 = \mathbf{0.4690}$ **Final Certainty:** 0.4690

Diagnosis: engine_temperature_sensor_lying

Statement: IF hard_cold_start = true AND CTS_reading_wrong = true THEN open_case(engine_temperature_sensor_lying) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.53, 0.65) = 0.53$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.53 \times 0.7 = \mathbf{0.3710}$ **Final Certainty:** 0.3710

Diagnosis: vapor_lock_ambush

Statement: IF hard_hot_start = true AND fuel_rail_pressure_drops = true THEN open_case(vapor_lock_ambush) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.9, 0.71) = 0.71$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.71 \times 0.7 = 0.4970$ **Final Certainty:** 0.4970

Diagnosis: radiator_strangled

Statement: IF coolant_temp > 110C AND radiator_flow_blocked = true THEN open_case(radiator_strangled) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.89, 0.84) = 0.84$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.84 \times 0.7 = 0.5880$ **Final Certainty:** 0.5880

Diagnosis: water_pump_stopped_pumping

Statement: IF coolant_temp > 110C AND water_pump_flow = 0 THEN open_case(water_pump_stopped_pumping) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.89, 0.5) = 0.50$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.50 \times 0.7 = 0.3500$ **Final Certainty:** 0.3500

Diagnosis: coolant_vanishing_internally

Statement: IF coolant_disappears AND no_external_leak = true THEN open_case(coolant_vanishing_internally) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.79, 0.77) = 0.77$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.77 \times 0.7 = 0.5390$ **Final Certainty:** 0.5390

Diagnosis: heat_circulation_broken

Statement: IF heater_output = cold AND coolant_level < min THEN open_case(heat_circulation_broken) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.71, 0.7) = 0.70$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.70 \times 0.7 = 0.4900$ **Final Certainty:** 0.4900

Diagnosis: heater_core_blocked_solid

Statement: IF heater_output = cold AND heater_core_flow = blocked THEN open_case(heater_core_blocked_solid) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.71, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = 0.4270$ **Final Certainty:** 0.4270

Diagnosis: fuel_and_ignition_conspiracy_confirmed

Statement: IF misfire = true AND O2_flat = true AND fuel_trim > 15% THEN close_case(fuel_and_ignition_conspiracy_confirmed) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.88, 0.85, 0.61) = 0.61$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.61 \times 0.7 = 0.4270$ **Final Certainty:** 0.4270

Diagnosis: engine_and_transmission_co_suspects

Statement: IF engine_light = ON AND transmission_temp_high = true AND rpm_drop = true THEN open_case(engine_and_transmission_co_suspects) **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.76, 0.61, 0.51) = 0.51$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.51 \times 0.7 = 0.3570$ **Final Certainty:** 0.3570

Diagnosis: engine_internal_destruction_imminent

Statement: IF oil_pressure_low = true AND knocking = true AND temp_high = true THEN close_case(engine_internal_destruction_imminent) **Total:** 76 rules **Calculation Steps:**

- **Step 1 (Evidence Logic):** $CF(\text{Evidence}) = \min(0.85, 0.62, 0.76) = 0.62$
- **Step 2 (Apply Rule):** $CF(H,E) = 0.62 \times 0.7 = 0.4340$ **Final Certainty:** 0.4340